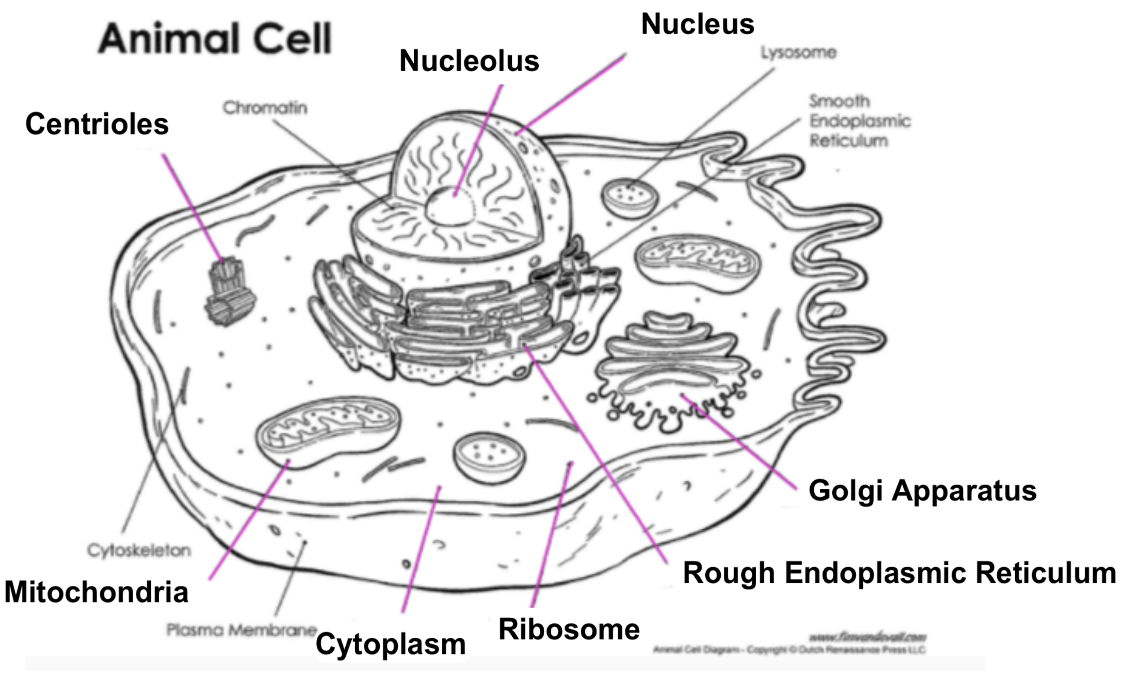
YEAR 10 SCIENCE
BIOLOGICAL SCIENCES
REVISION

1. Label the following simple cell drawing.

Nucleus Ribosome Mitochondria Rough Endoplasmic Reticulum
Golgi Apparatus Nucleolus Cytoplasm Centrioles



2. (a) Where in the cell do you find DNA?

Nucleus

(b) Describe the shape of the DNA molecule.

Twisted ladder made up of rung of bases.

(b) Name the ***4 bases*** in a DNA molecule.

- G Guanine
- A Adenine
- T Thymine
- C Cytosine

(b) Which bases are paired together in DNA?

G - C A - T

(c) Draw the adjoining strand of DNA for the following.

- G - T - T - A - C - A - G - T

- C - A - A - T - G - T - C - A

3. (a) Which process of cell division is responsible for the ***growth and repair of cells***?

Mitosis

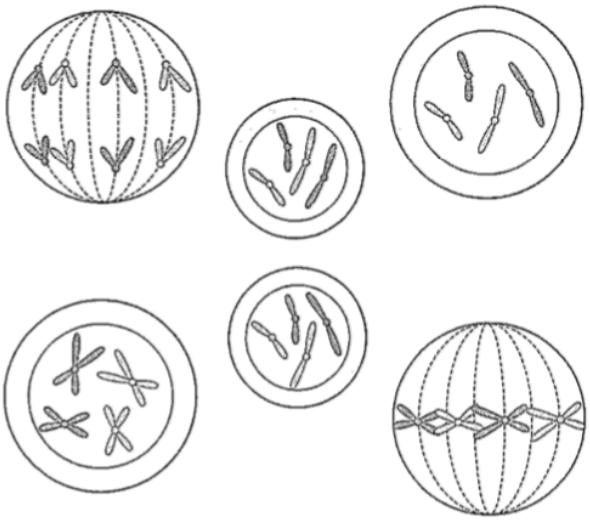
(b) Which process is involved in producing the ***reproductive cells*** of an animal?

Meiosis

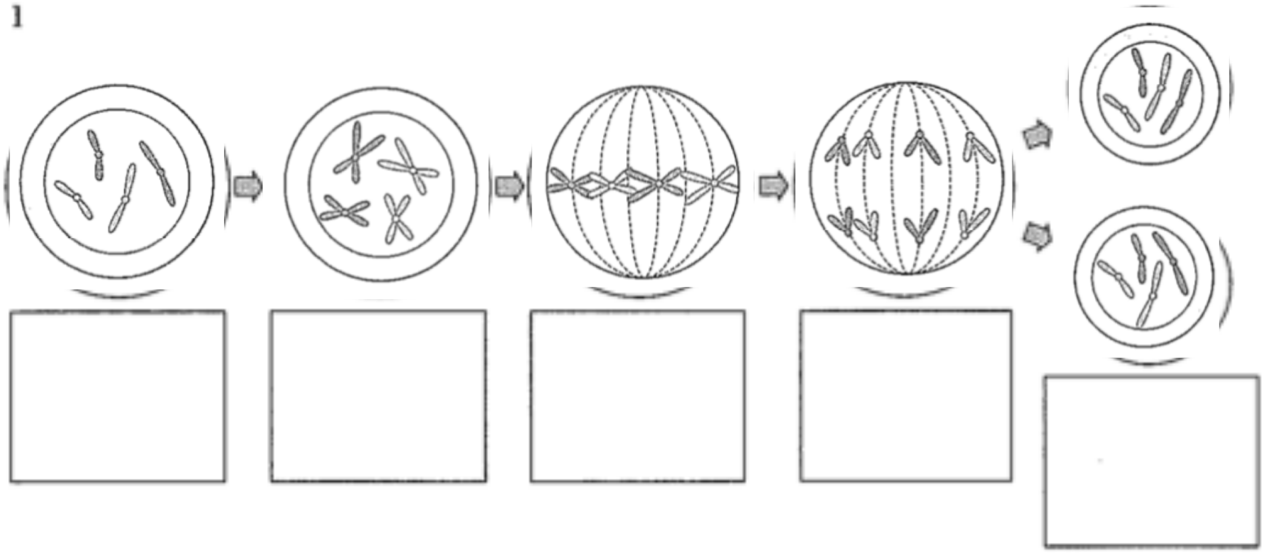
4. Define the following terms.

allele	One form of a gene e.g. T, t
pure-bred	Alleles the same e.g. TT, tt
hybrid	Alleles different e.g. Tt
homozygous	Alleles the same e.g. TT, tt
heterozygous	Alleles different e.g. Tt
genotype	Genes for a characteristic e.g. TT, Tt, tt
phenotype	How it looks e.g. tall, dwarf
dominant	Gene that masks (hides) another e.g. T (tall) masks t (dwarf)
recessive	Gene that is hidden and doesn't show e.g. t (dwarf)
haploid	Half the normal number of chromosomes
gene	Represents a trait in an organism e.g. b - blue eyes. Also called ***allele***.
incomplete dominance	A new trait shows e.g. RW is pink in some flowers (don't see red or white)
co-dominance	Both genes are dominant and show together e.g. BW - both show as blue and white

5. Below is a diagram showing the ***mitosis*** cell division process. The stages are in the wrong order. Try to put them in the correct order in the blank diagram.



1



6. (a) What are the genes (genotype) for the male and female sex chromosome pair?

XX - female XY - male

(b) Who determines the sex of a child - the mother or father? Explain.

Father - Y chromosome determines maleness

(c) A family has had five girls born into it, and the mother is pregnant with the sixth child. What is the chance that the child will be a girl?

50:50 (50%) Half of male sperm is X, half is Y

7. In humans, the ability to taste a chemical phenylthiocarbamide (PTU) is due to the presence of the ***dominant gene T***.

(a) Determine the genotype and phenotypes of the offspring produced by the marriage of a ***homozygous non-taster male (tt)*** and a ***heterozygous taster female (Tt)***. Set your working out.

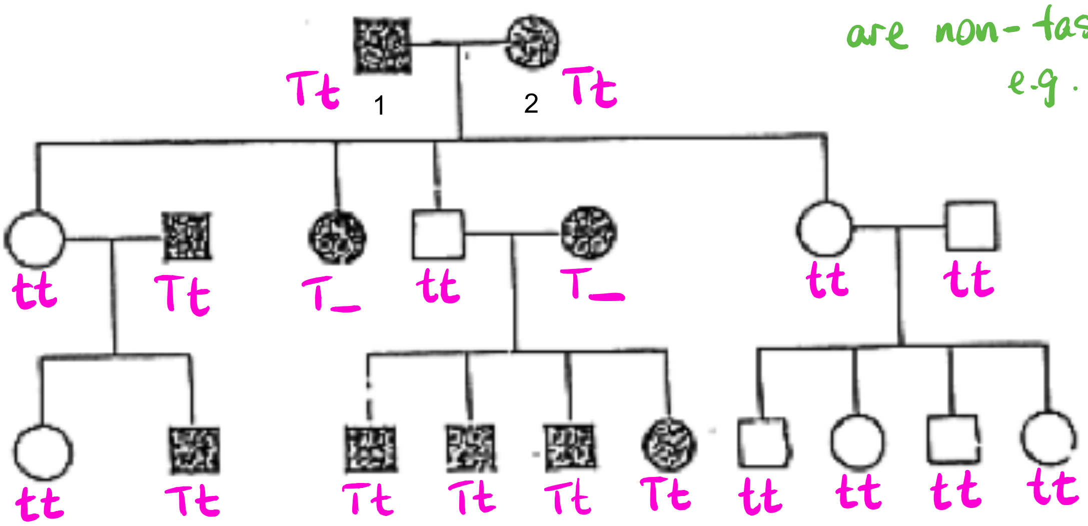
tt × Tt

♀ \ ♂	t	t
T	Tt	Tt
t	tt	tt

$\underbrace{\text{Tt Tt}}_{\text{50\% taster}}$ $\underbrace{\text{tt tt}}_{\text{50\% non-taster}}$

(b) Study the pedigree chart below showing the occurrence of tasters in a family. A shaded square or circle represents a taster.

All clear circles and squares are non-tasters. e.g. tt



(i) How many males are non-tasters? 4

(ii) How many females are tasters? 4

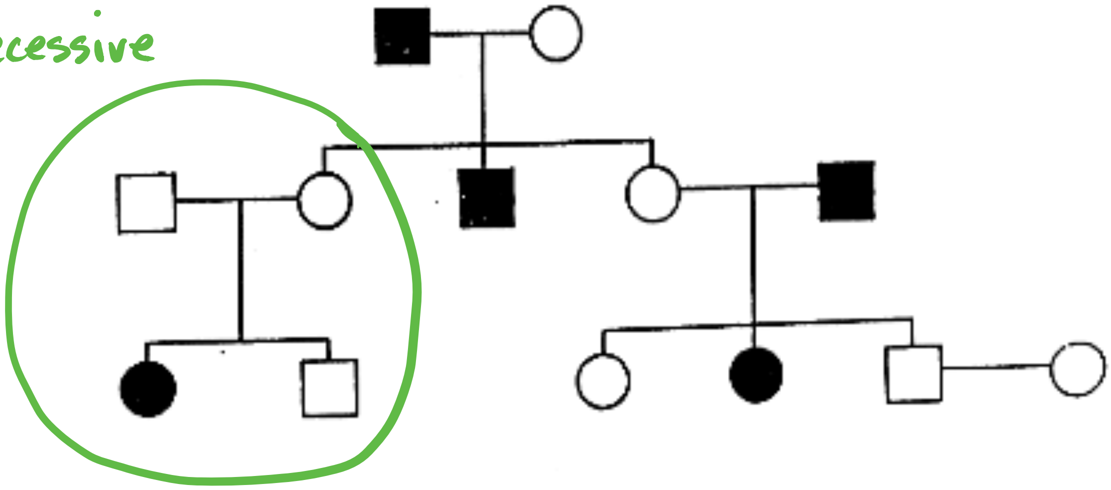
(iii) How many children did parents 1 and 2 have? 3

(iv) How many of their children are non-tasters? 2

(v) Try to write the genotype (actual genes) for each individual in the pedigree. ***Remember to work from the bottom up***.

8. The pedigree below shows the existence of straight hair in a family.

Recessive



Is straight hair a dominant or recessive trait? Explain your answer. You may need to circle part of the pedigree to help explain.

Recessive - two parents don't have the trait but one child does.